

TCX1004 Mathematical Techniques for Computing

Tutorial 1

Question 1

0) "If the input is valid, then the output is correct."

Ans : $q \rightarrow r$

1) "If the program compiles but the input is not valid, then the output is not correct."

Ans : $(p \wedge \neg q) \rightarrow \neg r$

2) "[The program compiles and the input is valid) or the function is efficient], and the algorithm does not terminate."

Ans : $[(p \wedge q) \vee s] \wedge \neg t$

3) "If the program compiles, then [either (the input is valid and the output is correct), or the algorithm does not terminate]."

Ans : $p \rightarrow [(q \wedge r) \vee \neg t]$

Question 2a(i)

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$	$\neg q$	$p \wedge \neg q$
true	true	true	false	false	false
true	false	false	true	true	true
false	true	true	false	false	false
false	false	true	false	true	false

Question 2a(ii)

p	q	$p \wedge q$	$\neg p$	$\neg q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
true	true	true	false	false	false	false
true	false	false	false	true	true	true
false	true	false	true	false	true	true
false	false	false	true	true	true	true

Question 2b) 1. It is not the case that it is raining or cold. $\neg(p \vee q)$

2. It is not raining or it is not cold. $\neg p \vee \neg q$

3. It is not raining and it is not cold. $\neg p \wedge \neg q$

4. It is not the case that it is both raining and cold. $\neg(p \wedge q)$

Statement 1 is logically equivalent to statement 3. Statement 2 is logically equivalent to statement 4.

Question 3

- 1) $\exists u \in U [\text{Orange}(u) \wedge \text{Diamond}(u)] - \text{True}$
- 2) $\forall u \in U [\text{Circle}(u) \rightarrow \text{Blue}(u)] - \text{False}$
- 3) $\forall s \in S, \exists d \in D [\text{Above}(d, s)] - \text{True}$
- 4) $\forall c \in C, \forall s \in S [\text{Above}(c, s)] - \text{True}$
- 5) $\exists u \in U, \forall c \in C [\text{Above}(u, c)] - \text{False}$
- 6) $\exists c \in C, \exists d \in D [\text{Grey}(c) \vee \neg \text{Orange}(d)] - \text{True}$

Question 4

1. $\forall x \in D, \exists y \in E [P(x, y) \wedge Q(y)]$
 Ans : $\neg(\forall x \in D, \exists y \in E [P(x, y) \wedge Q(y)])$
 $\exists x \in D (\neg(\exists y \in E [P(x, y) \wedge Q(y)]))$
 $\exists x \in D, \forall y \in E (\neg[P(x, y) \wedge Q(y)])$
 $\exists x \in D, \forall y \in E (\neg P(x, y) \vee \neg Q(y))$
2. $\exists x \in D, \forall y \in E [P(x, y) \rightarrow Q(y)]$
 Ans : $\neg(\exists x \in D, \forall y \in E [P(x, y) \rightarrow Q(y)])$
 $\forall x \in D (\neg(\forall y \in E [P(x, y) \rightarrow Q(y)]))$
 $\forall x \in D, \exists y \in E (\neg[P(x, y) \rightarrow Q(y)])$
 $\forall x \in D, \exists y \in E (P(x, y) \wedge \neg Q(y)) - \text{Lemma}$

Question 5

Part A

1. $\forall x \in \mathbb{N}, \exists y \in \mathbb{Z} [x=y] - \text{True}$
 Negation - $\exists x \in \mathbb{N}, \forall y \in \mathbb{Z} [x \neq y]$
2. $\forall p, q \in \mathbb{Q} [q=p] - \text{False}$
 Negation - $\exists p, q \in \mathbb{Q} [q \neq p]$
3. $\forall a, b \in \mathbb{Z} [a^2 = b^2 \rightarrow a=b] - \text{False}$
 Negation - $\neg(\forall a, b \in \mathbb{Z} [a^2 = b^2 \rightarrow a=b])$
 $\exists a, b \in \mathbb{Z} (\neg[a^2 = b^2 \rightarrow a=b])$
 $\exists a, b \in \mathbb{Z} ((a^2 = b^2) \wedge (a \neq b)) - \text{Lemma}$

Part B

1. $\forall a \in \mathbb{Z}, b \in \mathbb{Z} [a^2 \neq b^2 \rightarrow a \neq b] - \text{True}$
 Contrapositive - $\forall a \in \mathbb{Z}, b \in \mathbb{Z} [(a=b) \rightarrow (a^2 = b^2)]$
2. $\forall x \in \mathbb{Z} [(\forall y \in \mathbb{N} [x \neq y]) \rightarrow x \leq 5] - \text{True}$
 Contrapositive - $\forall x \in \mathbb{Z}, [(x > 5) \rightarrow \exists y \in \mathbb{N} (x \neq y)]$